

4

SEQUENTIAL LOGIC CIRCUITS

4.1. Sequential Logic Circuits

In sequential logic circuits, the output is a function of the present inputs as well as the inputs and outputs. Sequential circuit include memory elements to store past data. The flip-flop is a basic element of sequential logic circuits.

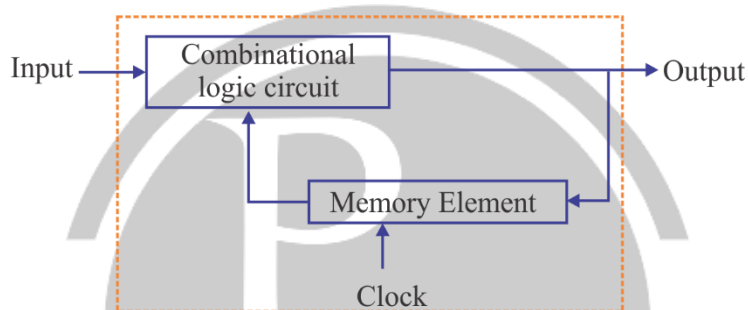


Fig. 4.1. General Block diagram of Sequential Logic Circuit

There are two types of sequential circuits:

4.1.1. Synchronous Circuits

The sequential circuits which are controlled by a clock are called synchronous sequential circuits. These circuits get activated only when clock signal is present.

4.1.2. Asynchronous Circuits

The sequential circuits which are not controlled by a clock are called asynchronous sequential circuits, i.e. the sequential circuits in which events can take place any time the inputs are applied are called asynchronous sequential circuits.

4.2. Difference Between Synchronous and Asynchronous Sequential Circuits

S.No.	Synchronous Sequential Circuits	Asynchronous Sequential Circuits
1.	In synchronous circuits, the change in input signals can affect memory elements upon activation of clock signal.	In asynchronous circuits, change in input signals can affect memory elements at any instant of time.
2.	In synchronous circuits, memory elements are clocked FF's.	In asynchronous circuits, memory elements are either unlocked FF's or time delay elements.
3.	The maximum operating speed of the clock depends on time delays involved.	Since the clock is not present, asynchronous circuits can operate faster than synchronous circuits.
4.	They are easier to design.	More difficult to design.

5.	Synchronous circuit	Asynchronous circuit
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4.3. Latches

Flip-flop is an electronic circuit or device which is used to store a data in binary form. Actually, flip-flop is a one-bit memory device and it can store either 1 or 0. Flip-flops is a sequential device that changes its output only when a clocking signal is changing. On the other hand, latch is a sequential device that checks all its inputs continuously and changes its outputs accordingly at any time independent of a clock signal. It refers to non-clocked flip-flops, because these flip-flops, because these flip-flops 'latch on' to a 1 or a 0 immediately upon receiving the input pulse.

4.3.1. General Block Diagram of a Latch or Flip-flop

Figure shown below is the general type of symbol used for a latch. In case of a flip-flop, a clock signal must be shown at input side. It has many inputs and two outputs, labelled Q and $\bar{Q}$. The Q output is the normal output of the latch and $\bar{Q}$ is the inverted output.

Note: A flip-flop is said to be in HIGH state or logic 1 state or SET state when $Q = 1$, and in LOW state or logic 0 state or RESET state or CLEAR state when $Q = 0$.

4.3.2. Difference between Latches and Flip-flops

S.No.	Latch	Flip-flop
1.	A latch is an electronic sequential logic circuit used to store information in an asynchronous arrangement.	A flip-flop is an electronic sequential logic circuit used to store information in a synchronous arrangement. It has two stable states and maintains its states for an indefinite period until a clock pulse is applied.
2.	One latch can store one-bit information, but output state changes only in response to data input.	One flip-flop can store one-bit data, but output state changes with clock pulse only.
3.	Latch is an asynchronous device and it has no clock input.	Flip-flop has clock input and its output is synchronised with clock pulse.
4.	Latch holds a bit value and it remains constant until new inputs force it to change.	Flip-flop holds a bit value and it remains constant until a clock pulse is received.
5.	Latches are level-sensitive, and the output tracks the input when the level is high. Therefore, as long as the level is logic level 1, the output can change if the input changes.	Flip-flops are edge sensitive. They can store the input only when there is either a rising or falling edge of the clock.

4.4. Latch

A latch is a type of bistable logic device or multivibrator that is most often used in applications that require data storage. The main characteristics of latch is that the output is not dependent solely on the on the present state of the input but also on the proceeding output state.

Latches are sometimes used for multiplexing data onto a bus. For example, data being input to a computer from a external source have to share the data bus with data from other sources. When the data bus becomes unavailable to external source, the existing data must be temporarily stored, and hence the latches are placed between the external source and data bus.

4.4.1. SR Latch

For the SR latch (S stands for set and R for reset). The logic circuit for SR latch is shown in figure below:

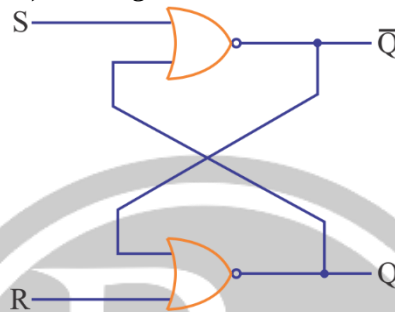


Fig. 4.1. Logic circuit of SR latch.

The state table for the SR latch is:

S	R	Q	Q^+	$\overline{Q^+}$
0	0	0	0	1
0	0	1	1	0
0	1	0	0	1
0	1	1	0	1
1	0	0	1	0
1	0	1	1	0
1	1	0	0	1
1	1	1	0	0

The symbol for SR Latch is:

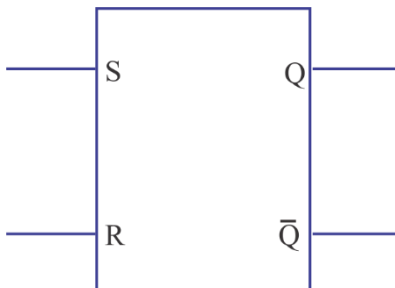


Fig. 4.2.

Obtaining the characteristic equations of the NOR gate based latch are; we get

$$Q^+ = \bar{R} \times S + \bar{R} \times Q = \bar{R} \times (S + Q) \text{ and } \bar{Q}^+ = \bar{S} \times R + \bar{S} \times \bar{Q} = \bar{S} \times (S + \bar{Q})$$

Note: It must be noted that the complementing Q^+ does not yield $\bar{Q}^+$.

Hence, the truth table for SR latch is

S	Q	Q^+	$\overline{Q}^+$	
0	0	Q	$\overline{Q}$	$\Rightarrow$ No change
1	1	0	1	$\Rightarrow$ Reset Q^+ to 0
1	0	1	0	$\Rightarrow$ Set Q^+ to 1
1	1	0	0	$\Rightarrow$ Forbidden state

However, the forbidden state ($S = R = 1$) is considered a don't care state.

Consider a Timing diagram for SR latch

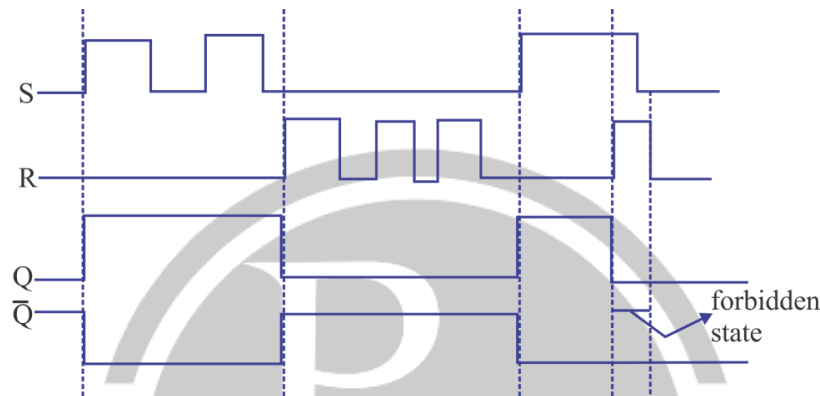


Fig. 4.3.

4.4.2. $\overline{S}\overline{R}$ Latch:

An $\overline{S}\overline{R}$ latch can be implemented using NAND gates, as shown in figure below

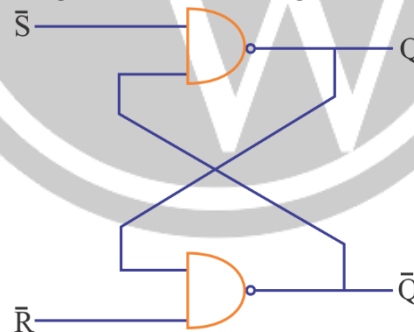


Fig. 4.4. Logic circuit for $\overline{S}\overline{R}$ Latch

The $\overline{S}\overline{R}$ latch is said to be set-dominant 1,
The symbol for $\overline{S}\overline{R}$ latch is shown below:

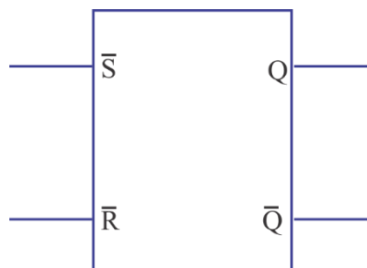


Fig. 4.5.

The truth table for $\overline{S}\overline{R}$ latch is given as:

$\overline{S}$	$\overline{R}$	Q^+	$\overline{Q}^+$	
1	1	Q	$\overline{Q}$	$\Rightarrow$ No change
1	0	0	1	$\Rightarrow$ Reset Q^+ to 0
0	1	1	0	$\Rightarrow$ Set Q^+ to 1
1	1	1	1	$\Rightarrow$ Forbidden state

Application of $\overline{S}\overline{R}$ latch: The application of $\overline{S}\overline{R}$ latch is in switch bouncing i.e. contact bounces of a push-button switch during its opening or closing can be eliminated by using $\overline{S}\overline{R}$ latch.

4.4.3. Gated SR Latch on Enable SR Latch or clocked SR Latch

A gated or level-sensitive SR latch uses a control signal C that can be used as a clock signal or can be used as enable input. The logic circuit diagram, symbol and truth is given as

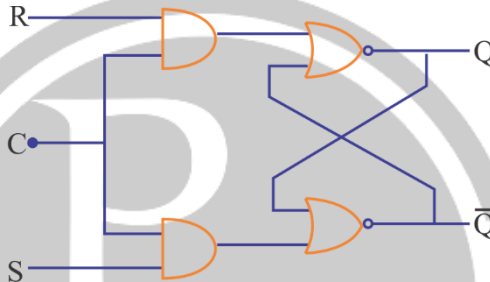


Fig. 4.6. Logic circuit of clocked SR latch.

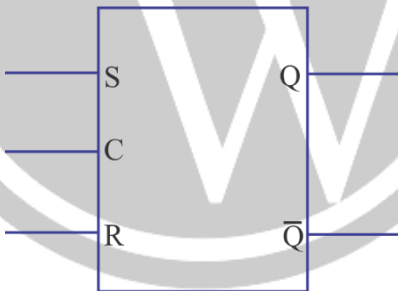


Fig. 4.7. Symbol for clocked SR latch

C	S	R	Q^+	$\overline{Q}^+$	
0	X	X	Q	$\overline{Q}$	} No change state
1	0	0	Q	$\overline{Q}$	
1	0	1	0	1	$\Rightarrow$ Reset
1	1	0	1	0	$\Rightarrow$ Set
1	1	1	0	0	$\Rightarrow$ Forbidden state

4.4.4. Gated $\overline{S}\overline{R}$ Latch or enable $\overline{S}\overline{R}$ Latch or clocked $\overline{S}\overline{R}$ Latch

Gated $\overline{S}\overline{R}$ latch is implemented using two NAND gates and an $\overline{S}\overline{R}$ latch. The logic circuit diagram, symbol and truth is given as

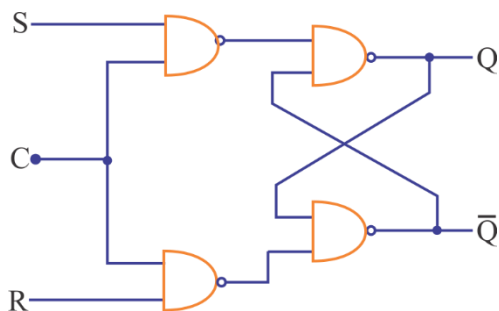


Fig. 4.8. Logic diagram of clocked $\bar{S}\bar{R}$ latch.

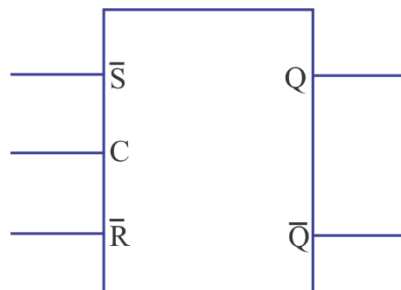


Fig. 4.9.

The truth table of the gated SR latch based on a $\bar{S}\bar{R}$ latch:

C	J	K	Q	$\bar{Q}^+$
0	X	X	Q	$\bar{Q}$
1	0	0	Q	$\bar{Q}$
1	0	1	0	1
1	1	0	1	0
1	1	1	1	1

The characteristic equation for SR flip-flop is given as

$$Q^+ = Q_{n+1} = S + \bar{R}Q_n = S + \bar{R}Q$$

4.5. Flip-Flops

Flip-flops are synchronous bistable devices also known as bistable multivibrator. Its output change its state only at a verified point (i.e. leading or trailing edge) on the triggering input called the clock (CLK), i.e. changes in the output occur in synchronization with the clock.

Flip-flops are edge-triggered or edge-sensitive whereas gated latches are level-sensitive.

4.5.1. Edge-triggered flip-flop

An edge-triggered flip-flop changes its state either at positive edge (rising edge) or at negative edge (falling edge) of the clock pulse.

There are two type of edge-triggered flip-flops. The key to identify an edge-triggered flip-flop is by its logic symbol by small triangle inside the block at the clock input C. This triangle is called the dynamic input indicator. Positive edge triggered has no bubble at input C whereas negative edge triggered has bubble at input C.

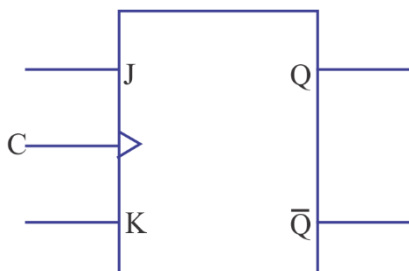


Fig. 4.10. Positive edge triggered flip-flop.

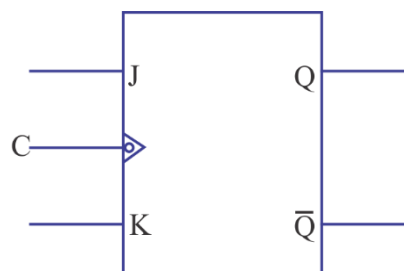


Fig. 4.11. Negative edge-triggered flip-flop.

4.5.2. Basic JK flip-flop

JK Flip-flop (J as a set input and K as a reset input) is the most versatile of the basic flip-flops.

The logic circuit of the gated JK flip-flop is shown in figure below:

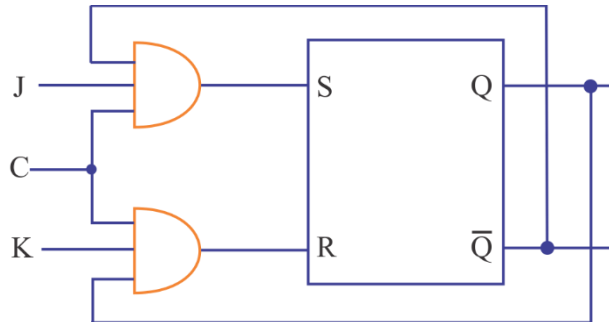


Fig. 4.12. Logic circuit diagram of clocked JK flip-flop.

The state table for the JK flip-flop is given as

C	J	K	Q	Q ⁺
0	X	X	X	Q
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

Hence, the truth table becomes,

C	S	R	Q ⁺	$\bar{Q}^+$	
0	X	X	Q	$\bar{Q}$	} No change state
1	0	0	Q	$\bar{Q}$	
1	0	1	0	1	⇒ Reset
1	1	0	1	0	⇒ Set
1	1	1	0	0	⇒ Forbidden state

Note: The forbidden state, inherent to SR flip-flop is eliminated by adding two feedback loops such that the output becomes 1 only if Q = 0 and reset to only if Q = 1.

It should also be noted that when the inputs (J & K) are set to 1 and clock signal change to 1, then the feedback value of Q & $\bar{Q}$ forced the flip-flop is toggle its value.

(i.e. to switch its state to its logical complement) hence, to ensure this operation in smooth fashion, the pulse width of the clock must be smaller than the propagation delay of the flip-flop.

The characteristic equation of the JK flip-flop

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n \text{ or } Q^+ = J\bar{Q} + \bar{K}Q$$

4.5.3. T-flip-flop

A JK flip-flop can be transformed into a T- flip-flop (T stands for Toggle). When T flip-flop is activated, its output changes its state at every time a pulse is applied to the input T.

The logic circuit of the gated T flip-flop is.

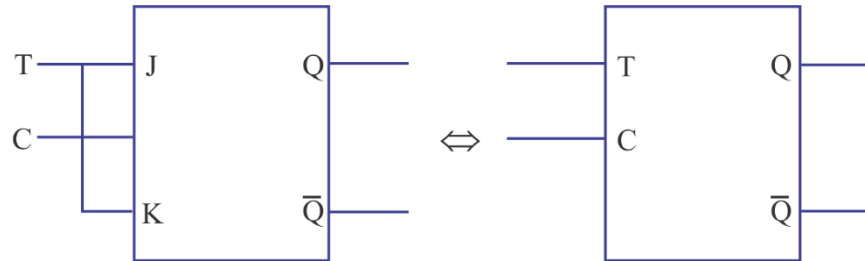


Fig. 4.13.

The state or characteristic table for T flip-flop is

C	T	Q	Q ⁺
0	X	X	Q
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

As $J = K = T$, we obtain the characteristic equation as

$$Q^+ = T \times \bar{Q} \times C + (\bar{T} + \bar{C}) \times Q$$

If $C = 1$, the characteristic the equation is reduced to

$$Q^+ = T \oplus Q$$

$$\text{If } C = 0, Q^+ = Q$$

Hence, the truth table of the T-flip flop is given as

C	T	Q ⁺	$\bar{Q}$
0	X	Q	$\bar{Q}$
1	0	Q	$\bar{Q}$
1	1	$\bar{Q}$	

} ⇒ No change state
⇒ Toggle

Fig. 4.14.

4.5.4. D Flip-Flop

D-flip-flop can be obtained by use of only two combinations of S-R or J-K flip-flop. It has only one input i.e. D-input or data input.

The logic symbol for D- flip-flop is given as

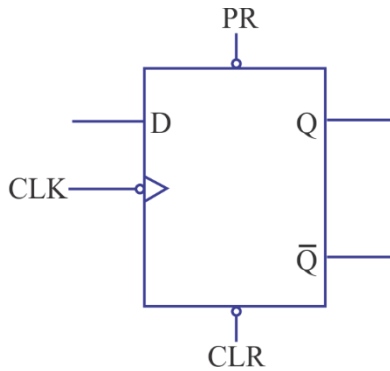


Fig. 4.15.

The truth table for D-flip-flop is

Input	Output
D	Q_{n+1}
0	0
1	1

The characteristic equation of D-flip-flop is:

$$Q_{n+1} = D$$

4.5.5. Excitation table of Flip-flops

The truth table of a flip-flop is sometimes referred as characteristic table as it specifies the operational characteristics of the flip-flop there may occurs some situations in which the present state and the next state of the circuit is desired and known. Then the designing of input conditions to as to fulfil the requirements of the circuit, there is a table called excitation table. It is very important and useful design aid for sequential circuit.

The excitation table for flip-flops:

Present state	Next state	SR Flip-flop		JK Flip-flop		T Flip-flop	D Flip-flop
		S	R	J	K	T	D
0	0	0	×	0	×	0	0
0	1	1	0	1	×	1	1
1	0	0	1	×	1	1	0
1	1	×	0	×	0	0	1

4.6. Operating Characteristics of Flip-Flops

4.6.1. Propagation Delay Time:

Propagation delay time is the time interval required after an input signal has been applied for the resulting output change to occur.

There are four categories of propagation delay times which are as follows:

A. Propagation delay t_{PLH} , it is measured from the triggering edge of the clock pulse to Low-to-High transition of the output.

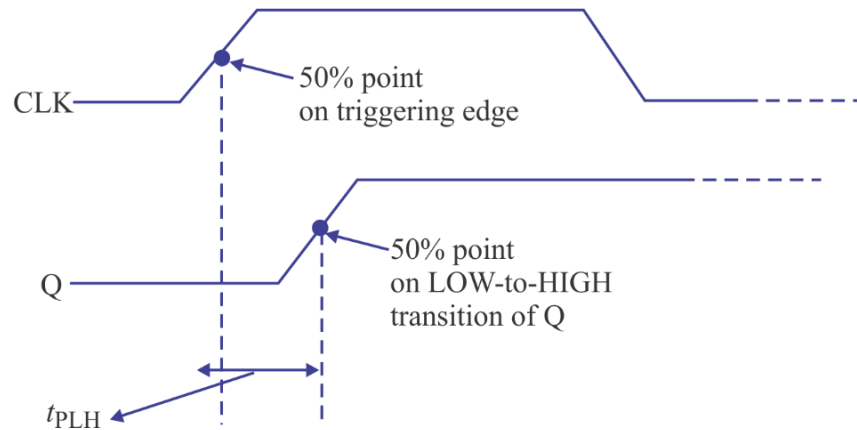


Fig. 4.16.

B. Propagation delay t_{PLH} , it is measured from the triggering edge of the clock pulse to HIGH-to-LOW transition of the output.

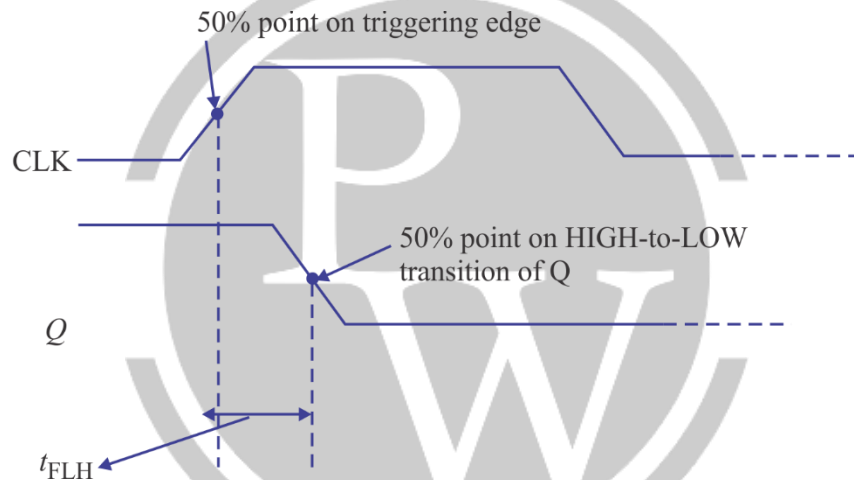


Fig. 4.17.

C. Propagation delay t_{PLH} , it is measured from the leading edge of the PRESET input to LOW-to-HIGH transition of the output.

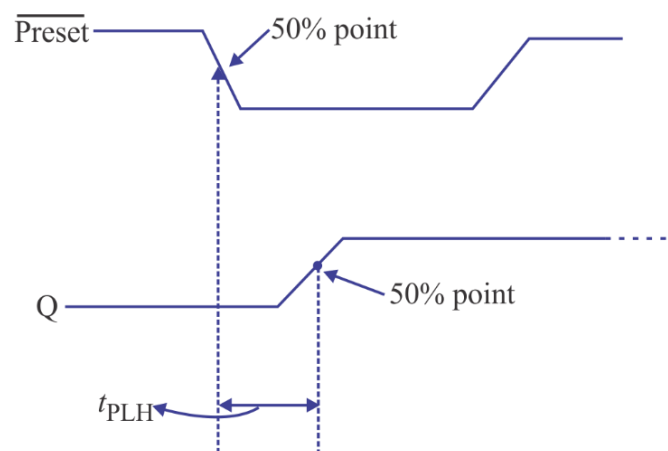


Fig. 4.18.

D. Propagation delay t_{PHL} , it is measured from the leading edge of the clear input to the HIGH-to-LOW transition of the output.

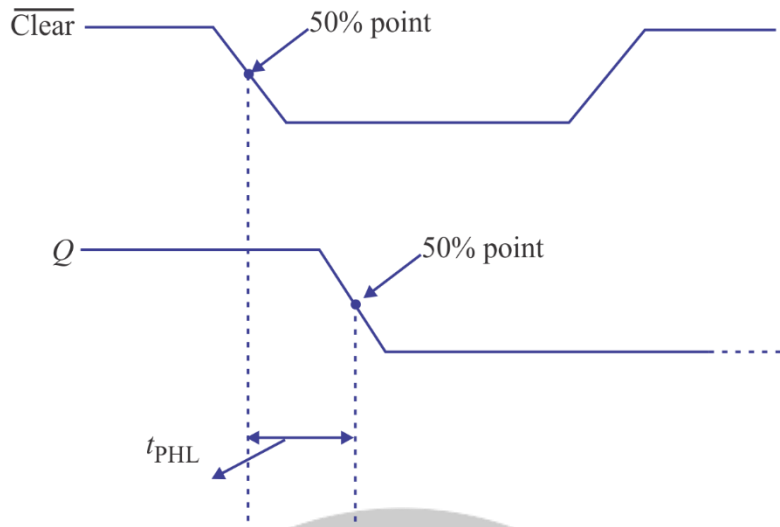


Fig. 4.19.

4.6.2. Set-up time (t_s)

It is the minimum time interval required for the logic levels (0 or 1) to be maintained constantly on the inputs (J, K or D) prior to the triggering edge of the clock pulse in order for the levels to be reliably clocked into the flip-flop.

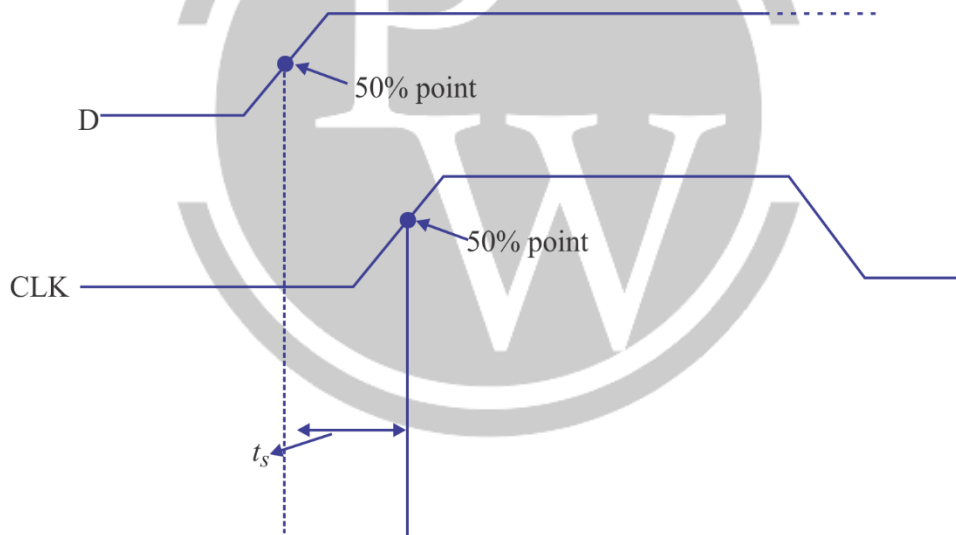


Fig. 4.20.

4.6.3. Hold time (t_h)

It is the time for which the data must remain stable after the triggering edge of the clock.

4.6.4. Clock-pulse width

The minimum time duration for which the clock pulse must remain HIGH and LOW which are designed by manufacturers. Failure to clock pulse width results in unreliable triggering.

4.6.5. Maximum clock frequency

The maximum clock frequency (f_{max}) is the highest rate at which flip-flop can be reliably operated.